

Heterogeneous Effects of Dirty Wind on Emergency Medical Service Demand: A Causal Forest Analysis

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Introduction & Background

Urban air pollution and health:

- Air pollution is a major public health challenge in dense urban settings.
- New York City** as a case study:
 - The city's population density, reliance on transportation, and neighborhood-level inequalities create uneven exposure and vulnerability.
 - Studying NYC offers insights into how environmental stressors interact with **socioeconomic disparities**.

Wind direction as natural variation:

- Southwest (SW) winds** often carry higher pollutant loads (“dirty wind”).
- This variation provides a quasi-experimental setting to identify **causal effects** of pollution on EMS calls using **causal forest**.

Research questions:

- Does exposure to dirty SW wind increase EMS demand?
- Which neighborhoods are most susceptible, and what traits define them?
- How do results differ when Manhattan is included versus excluded?

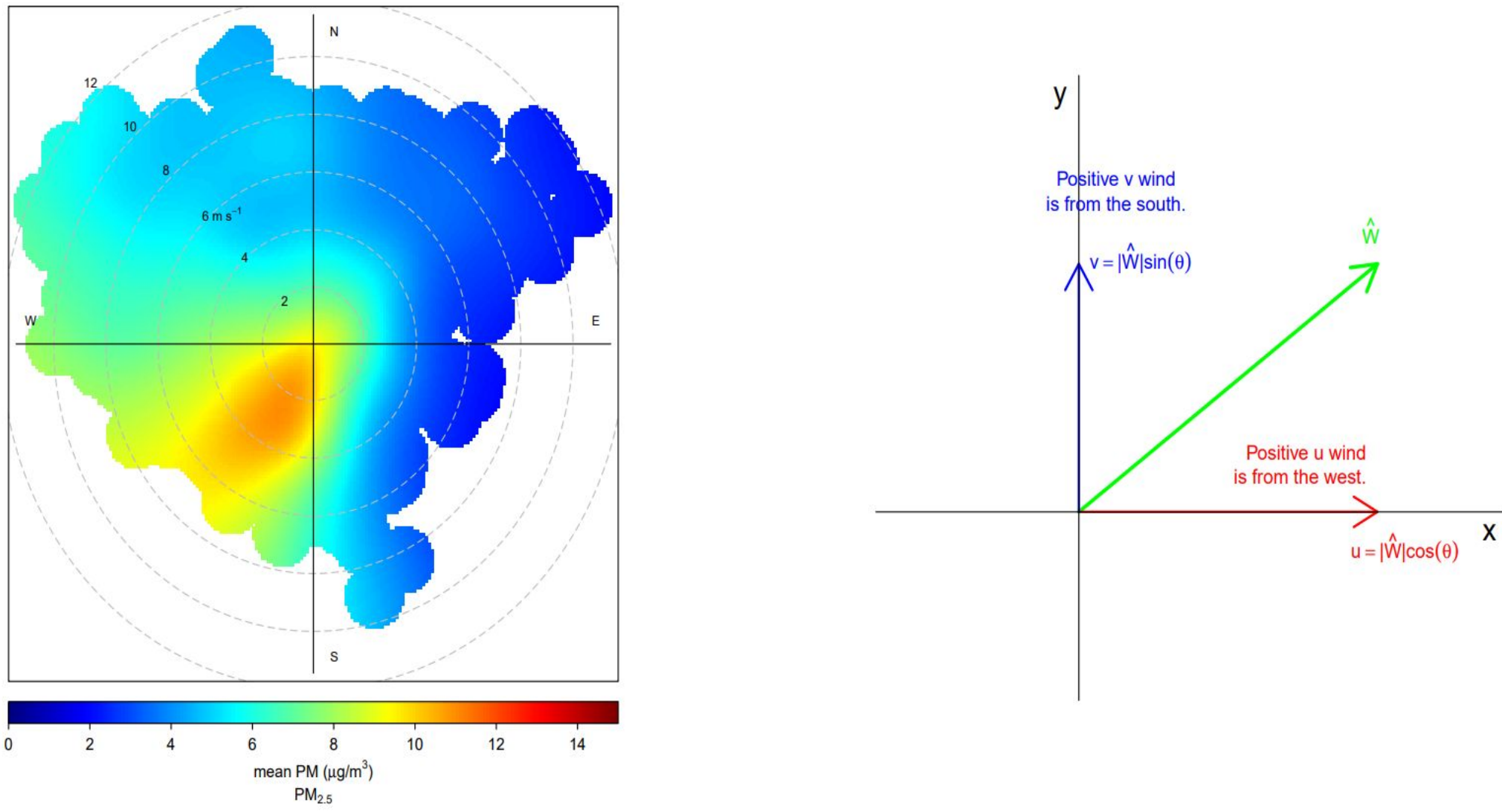


Fig 1: Wind Direction and Air Pollution

Data

Use Monthly ZIP Code Tabulation Area (ZCTA)–level data on:

- EMS calls** (respiratory-specific and total) [Y].
- Population and household traits (ACS)** [X].
- Wind direction:** Treatment variables (continuous, binary, quantile) [Wind].

Sample Size	$N = 12,328$ (with Manhattan); $N = 9,736$ (without Manhattan).
Outcome ^a	Respiratory count rate; Total EMS call rate.
Treatment ^b	<i>WindSW (continuous)</i> : continuous share/intensity of SW winds; <i>WindSW (binary)</i> : indicator above/below a threshold; <i>WindSW (quantile)</i> : categorical bins of SW wind share.
Covariates	
<i>Wealth</i>	Income inequality (Gini index); % of households below poverty line.
<i>Education</i>	% with bachelor's degree; % with master's degree; % with doctorate; % with high school diploma; % with GED; % with associate's degree.
<i>Language</i>	% of residents with limited English proficiency.
<i>Race/Ethnicity</i>	% nonwhite; % Hispanic; % Black; % Asian.
<i>Life Quality</i>	% of households receiving food stamps; % without health insurance; % unemployed; % foreign-born.
<i>Household</i>	% with rent burden; % living in overcrowded rental units; % without a vehicle; % with disability; % female single-parent households; % male single-parent households.
<i>Age</i>	% of population aged 65 or older.

^a Rates are computed as *counts per 1,000,000* for comparability and numerical stability.

^b Wind direction is measured as the monthly average direction in radians. “SW wind” corresponds to $3.14 \leq \text{wind_dir_avg} \leq 4.71$.

Table 1: Data & Variable Summary

Method

CART & Random Forest

- In Statistical learning, classification and regression trees (**CART**) are commonly used for prediction and correlation tasks.
- Causal tree** uses similar algorithm but measuring **Conditional Average Treatment Effects (CATEs)** among each population group.
- Causal Forests(CF)** grow many honest causal trees (with bootstrapping, random feature selection, and sample-splitting) to estimate treatment effect heterogeneity.
- They aggregate leaf-level CATEs across trees, providing stable estimates of how treatment effects vary with covariates.
- We built 12 CFs using different combinations of Data, Y and Wind.

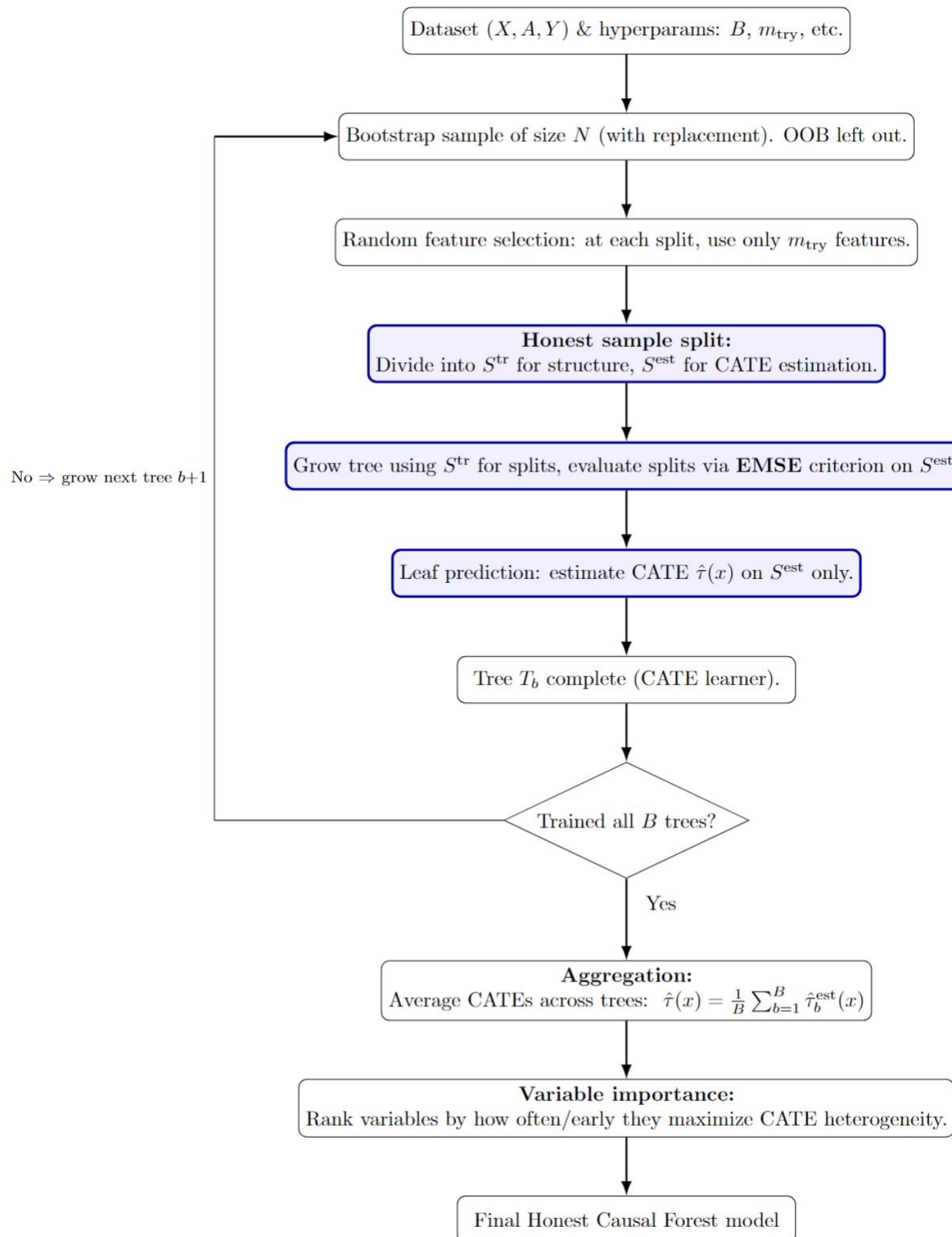
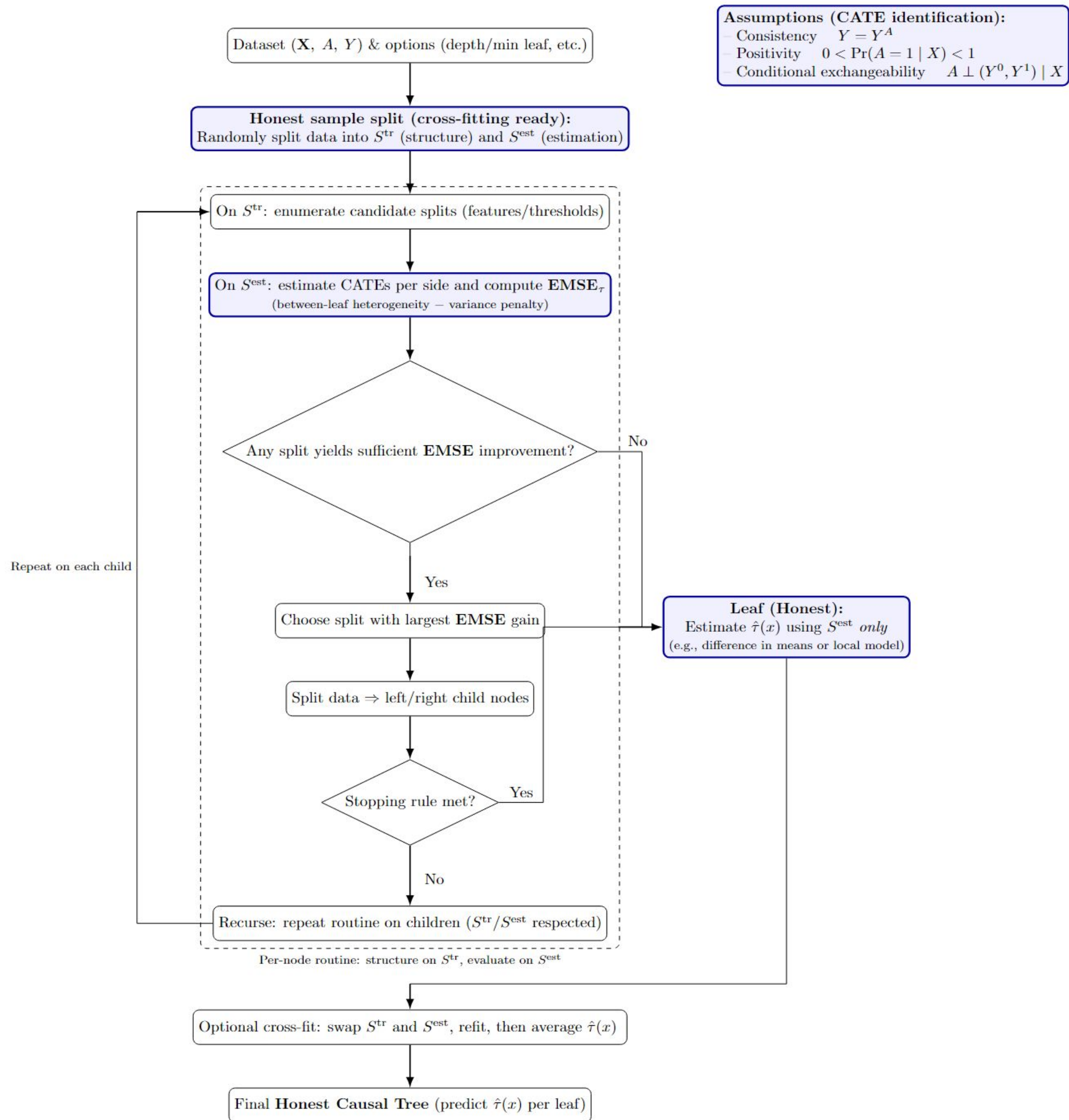


Fig 2&3: Honest Causal Tree and Causal Forest

Result

Causal Forest Visualization

- The outputs of causal forest are Conditional Average Treatments (CATEs) of each individual - **Each Month For Each Zip Code**.
- CATE: The expected difference in outcomes between treated and untreated groups, given a specific set of covariates X.

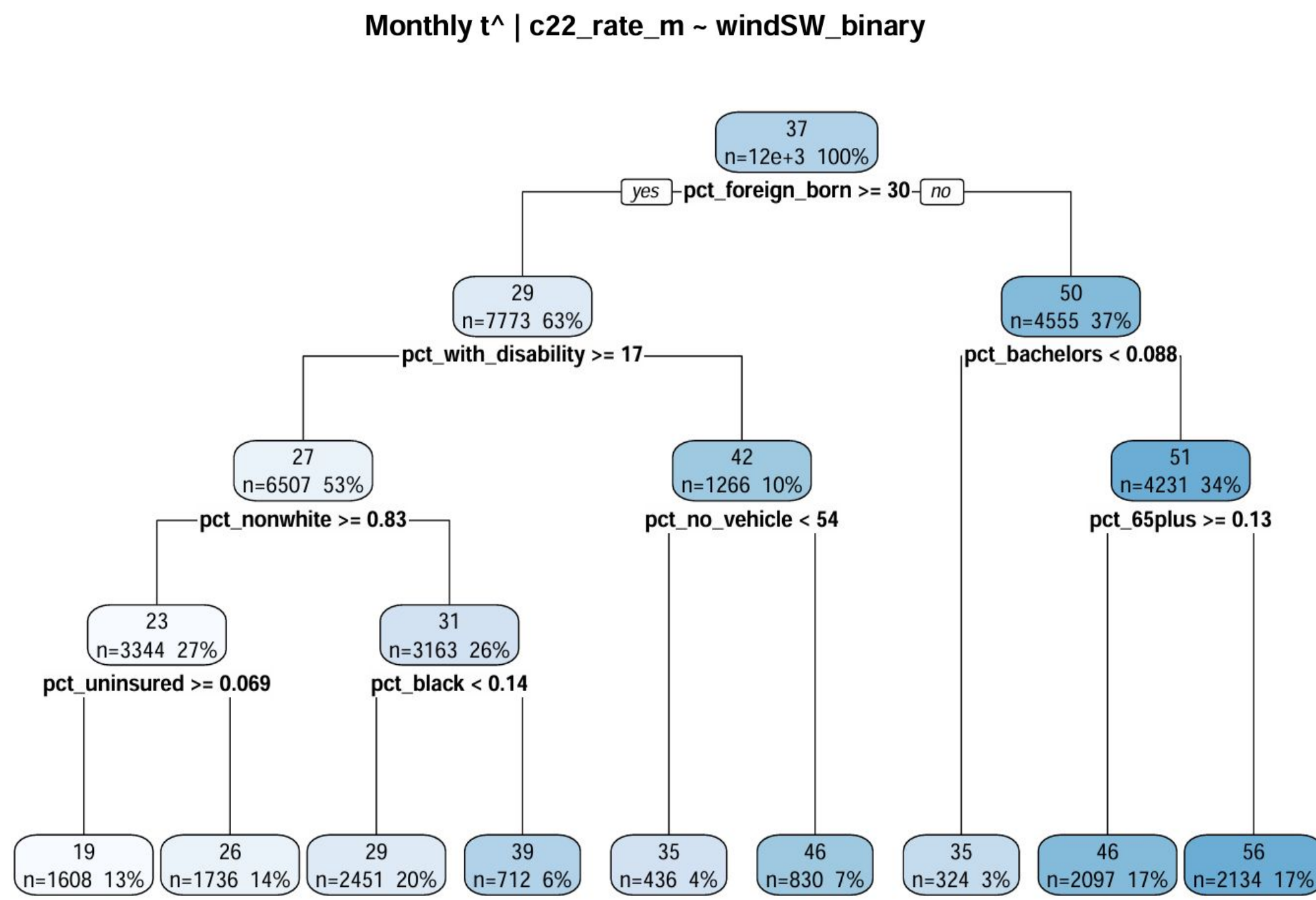


Fig 4: Example Random Forest CATEs Visualization Using Tree

Comparison Without Manhattan

- After running models on all NYC, we found that the most susceptible subpopulations have traits such as **low rent burden**, **high income inequality** and **low vehicle usage**, which are traits common for **Manhattan Population**.
- In order to make sure whether Manhattan is a special outlier, we repeat the models on data set without Manhattan.
- Across both data set, race/ethnicity (**low % nonwhite**, **higher % Black share**, **low foreign-born share**) and socioeconomic factors (**low rent burden**, **high inequality**, **low vehicle access**, **food-stamp use**, **single-parent households**) repeatedly define subpopulations most harmed by SW winds.

Trait	With Manhattan (4 models)			
	c22 cont	c22 bin	tot cont	tot bin
High inequality (Gini)	✓			
Low rent burden	✓			
Low foreign-born share	✓	✓		
Lower % nonwhite	✓	✓		
Higher % Black share	✓	✓		
Low elderly (65+)		✓		
High education (High bachelor's)	✓	✓		
High food-stamp usage			✓	
High female single-parent hlds			✓	
Low vehicle access (no vehicle)	✓			✓
Trait	Without Manhattan (4 models)			
	c22 cont	c22 bin	tot cont	tot bin
High inequality (Gini)	✓			
Low rent burden	✓			
Low foreign-born share	✓	✓		
Lower % nonwhite	✓	✓		
Higher % Black share	✓	✓		
Low elderly (65+)				
Low education (Low bachelor's)				✓
High food-stamp usage		✓	✓	
High female single-parent hlds			✓	
Low vehicle access (no vehicle)	✓			✓

Notes: Traits listed are those that define subpopulations most harmed by dirty SW wind in causal forest tree leaves. “c22” = respiratory EMS rate; “tot” = total EMS rate; cont/bin = windSW continuous/binary.

Table 2: Comparison of Models Built on 2 Data Set

Reference:

Brooks, John M., Cole G. Chapman, Sarah B. Floyd, Brian K. Chen, Charles A. Thigpen, and Michael Kissenberth. 2022. “Assessing the Ability of an Instrumental Variable Causal Forest Algorithm to Personalize Treatment Evidence Using Observational Data: The Case of Early Surgery for Shoulder Fracture.” *BMC Medical Research Methodology* 22 (1): 190.

Conte, Marc N., Matthew Gordon, Nicole A. Swartwood, Rachel Wilwending, and Chu A. Alex Yu. 2024. “Observational Studies Generate Misleading Results about the Health Effects of Air Pollution: Evidence from Chronic Air Pollution and COVID-19 Outcomes.” *PloS One* 19 (1): e0296154.

Davis, Jonathan M. V., and Sara B. Heller. 2017. “Using Causal Forests to Predict Treatment Heterogeneity: An Application to Summer Jobs.” *American Economic Review* 107 (5): 546–50.

Herrnstadt, Evan, Anthony Heyes, Erich Muehlegger, and Sodeeh Saberian. 2021. “Air Pollution and Criminal Activity: Microgeographic Evidence from Chicago.” *American Economic Journal. Applied Economics* 13 (4): 70–100.

Jawadekar, Neal, Katrina Kezios, Michelle C. Odden, Jeanette A. Stingone, Sebastian Calonico, Kara Rudolph, and Adina Zeki Al Hazzouri. 2023. “Practical Guide to Honest Causal Forests for Identifying Heterogeneous Treatment Effects.” *American Journal of Epidemiology* 192 (7): 1155–65.

Miller, Steve. 2020. “Causal Forest Estimation of Heterogeneous and Time-Varying Environmental Policy Effects.” *Journal of Environmental Economics and Management* 103 (102337): 102337.

Venkatasubramaniam, Ashwini, Bilal A. Mateen, Beverley M. Shields, Andrew T. Hattersley, Angus G. Jones, Sebastian J. Vollmer, and John M. Dennis. 2023. “Comparison of Causal Forest and Regression-Based Approaches to Evaluate Treatment Effect Heterogeneity: An Application for Type 2 Diabetes Precision Medicine.” *BMC Medical Informatics and Decision Making* 23 (1): 110.

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